

Low-tech graphite sensor

Main features

- ❖ Low power usage (3.3V-5V)
- ❖ Low cost
- ❖ Eco-friendly
- ❖ Easy to use
- ❖ Ergonomic and portable
- ❖ Easily repairable

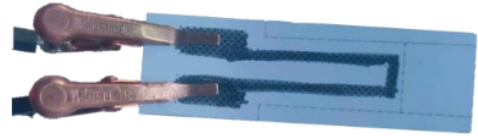


Figure 1 : Graphite sensor

Applications

- ❖ Tests findings of *Pencil Drawn Strain Gauges and Chemiresistors on Paper*
- ❖ Applied using a transimpedance amplifier connected to the ADC of an Arduino Uno Card
- ❖ Pedagogical tool for students to create and implement their own PCB design

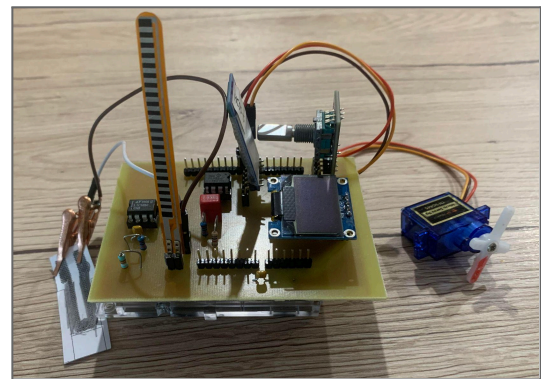


Figure 2 : Graphite sensor connected to the PCB

General description

This innovative sensor, conceptualized and made by students from Engineering Physics Department of INSA Toulouse, is a tool inspired by the publication *Pencil Drawn Strain Gauges and Chemiresistors on Paper*¹.

This research paper provides a simple, cost-efficient, and highly pedagogical tool for students to master their skills in Physics, Electronics, Programing, and Sensor Design. The sensor presented in the publication is a small piece of paper coated with a graphite layer from a pencil.

¹ by LIN Cheng-Wei, ZHAO Zhibo, KIM Jaemyung et HUANG Jiaying, 2014. Pencil Drawn Strain Gauges and Chemiresistors on Paper. Scientific Reports. 22 janvier 2014. Vol.4, n°1, pp.3812. DOI 10.1038/srep03812.

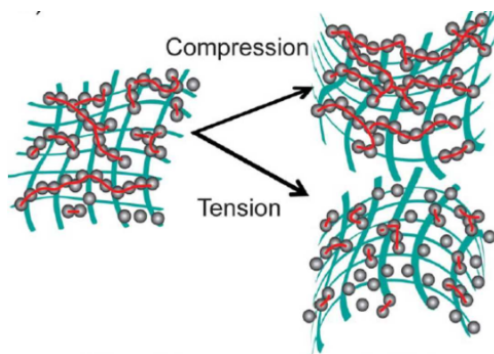


Figure 3 : Compression & tension in a granular system

Due to the deposited graphite on the piece of paper, the electrons are able to move freely from particle to particle due to quantum tunneling. This effect is extremely sensitive to the slightest movement of the paper. We observe that compressing or stretching the graphite will change the resistivity and the conductivity of the sensor. Thanks to this, we will be able to trace the deformation like a traditional strain gauge.

The structure of the graphite layer depends on the type of pencil used. We conducted tests with different types of pencils.

To achieve this, a transimpedance amplifier is needed to amplify the signal and then calculate the deformation. The transimpedance amplifier, the sensor and the other components are mounted on a PCB and set on an Arduino Uno card.

Dimensional and electrical diagram

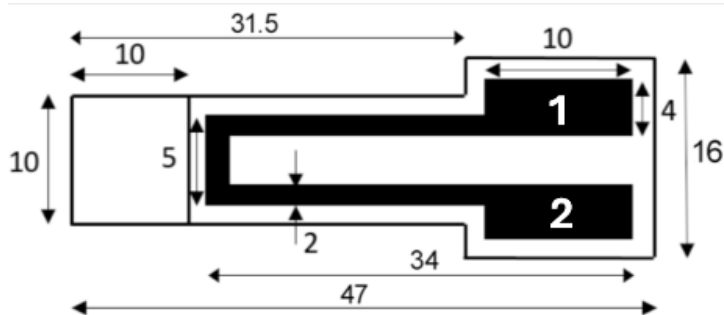


Figure 4 : Top view - Dimensions (mm)

■ Graphite layer

Thickness : 0.2mm

Pin	Symbol	Parameter
1	Vcc	Supply voltage
2	Vout	Out sensor

Table 1 : Specifications of the electrical diagram

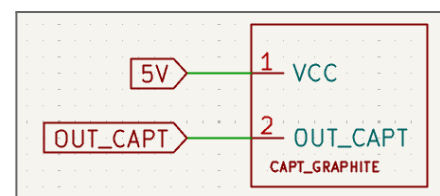


Figure 5 : Sensor's schematic

Note : Vcc is typically a +5V voltage. Both pins are reversible.

Technical specifications

Sensor	Strain sensor
Sensor Type	Passive (5V power supply required)
Materials	Graphite (4B to 2H pencils)
Nature of output signal	Analog
Nature of measurand	Voltage
Typical response time	< 50 ms
Typical use	Evaluation of a compression/tension deformation

Table 2 : Technical specifications of the graphite sensor

Absolute maximum ratings

Parameter	Symbol	Unit	Value		
			Min	Typical	Max
Supply voltage	Vcc	V	-	5.0	-
Temperature	T	°C	10	20	30
Air humidity	-	%	30	-	60
Air quality	-	%N2/O2	-	80/20	-
Life cycle	-	-	10	-	15
Paper thickness	-	mm	0.15	-	0.3
Pencil tone ²	-	-	4B	-	2H

Table 3 : Absolute maximum ratings of the graphite sensor

² Corresponds to the US grading system.

Electrical characteristics

Parameter	Unit	Value		
		Min	Typical	Max
Power supply	V	3.0	5.0	7.0
HB	k Ω	185	385	585
4B	k Ω	25	37.5	50
5B	k Ω	55	95	135

Table 4 : Values of the sensor's resistance depending on the type of pencil

Note : Values highly depend on the amount of graphite deposited.

Performance characteristics

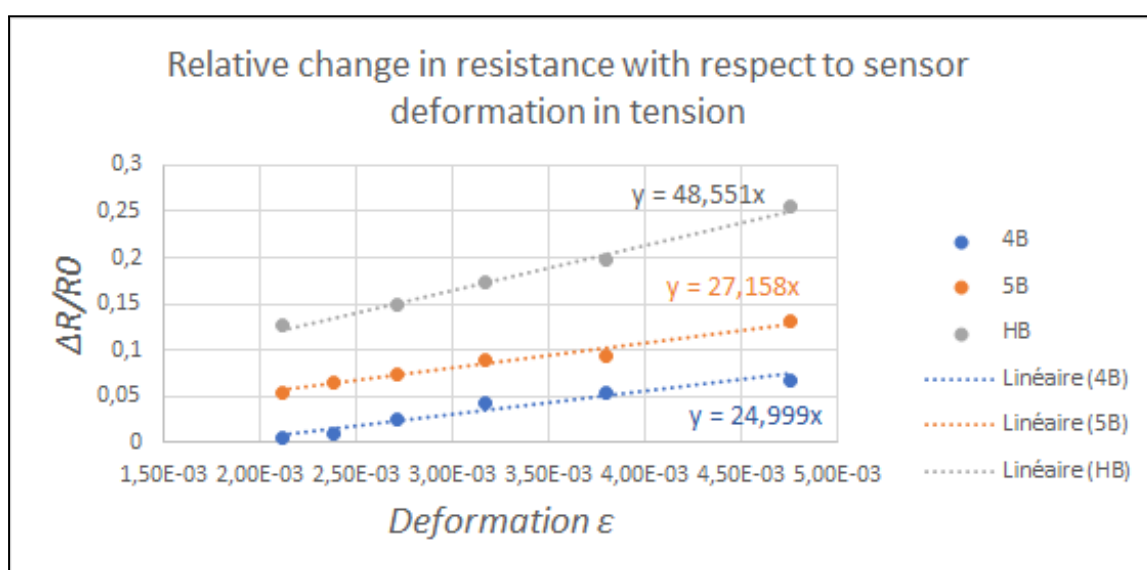


Figure 6 : Performance characteristics for tension

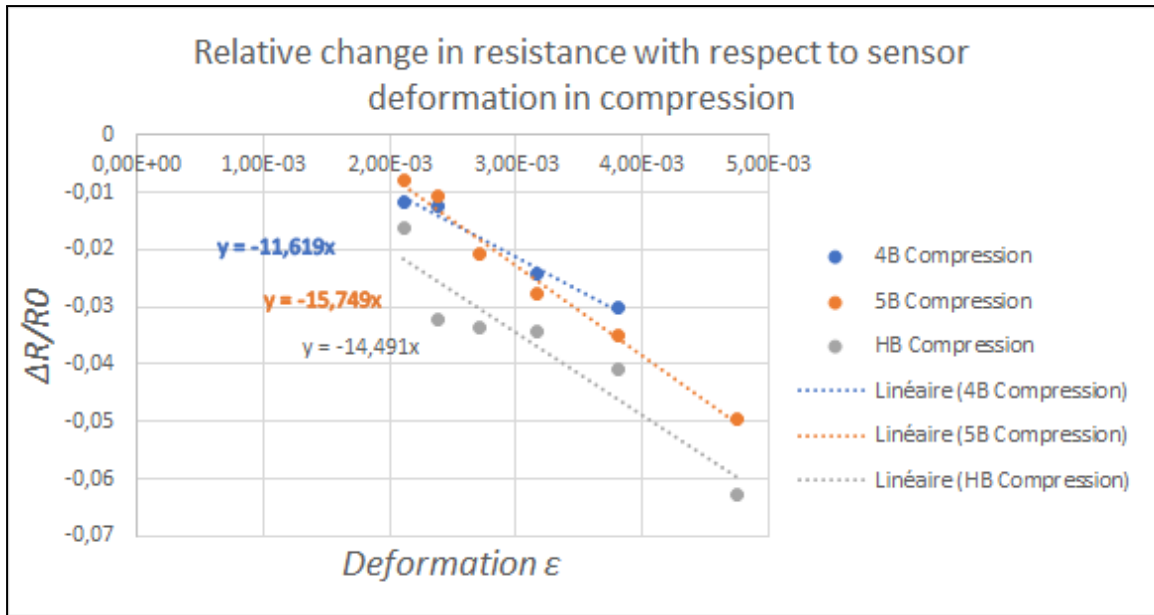


Figure 7: Performance characteristics for compression

Applications

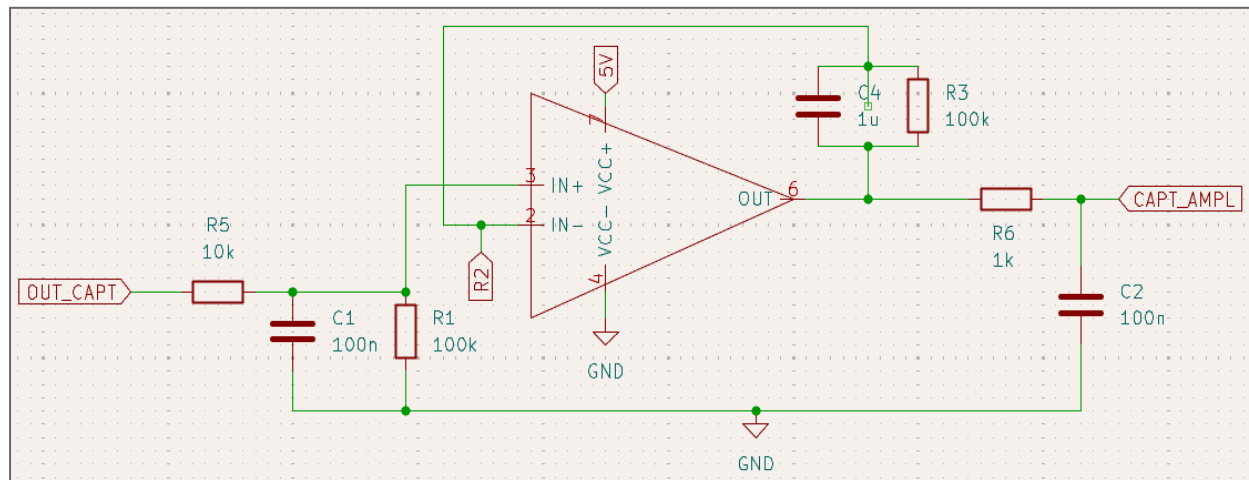


Figure 8: Transimpedance amplifier circuit

The output signal of the sensor is transmitted to an Analog-to-Digital (ADC) converter on the Arduino Uno Card. We have to make sure that the signal received by the ADC is big enough to be read but does not saturate. Therefore, the sensor is connected to a transimpedance amplifier circuit.

The circuit is equipped with a low-pass and high-pass filters to cancel out the noise from the sector, the current and the amplification of the signal.

On our PCB, we have installed a digital potentiometer in lieu of the R_2 resistance in order to match the amplification of the circuit for each pencil tone.

We can calculate the value of the resistance of the sensor knowing the value of the final voltage, with the following formula :

$$R_{sensor} = R_1 \left(1 + \frac{R_3}{R_{variable}} \right) \frac{V_{cc}}{V_{adc}} - R_1 - R_5$$